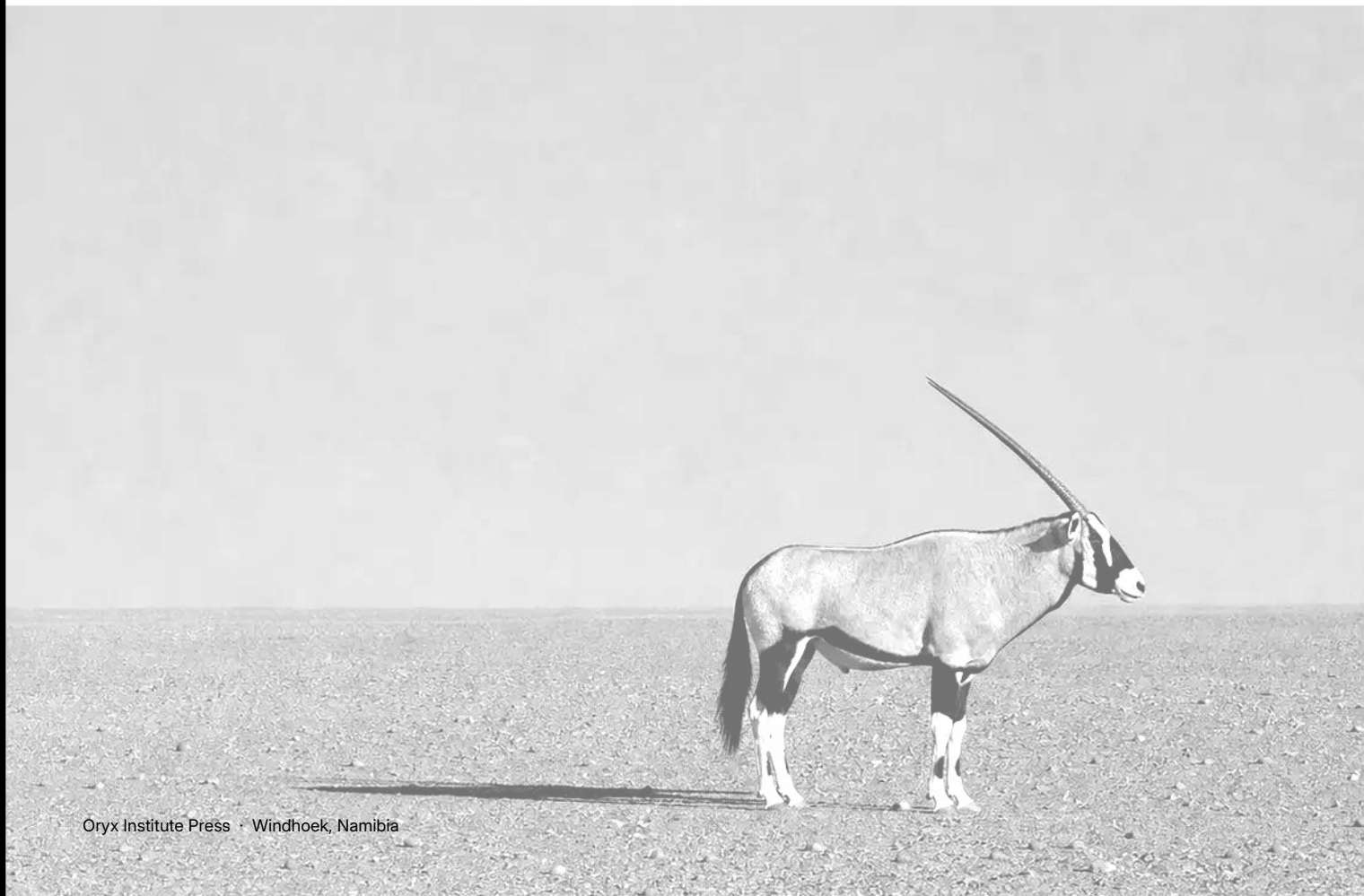
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TEXTBOOK · LEVEL 5

Foundations of Applied Measurement

A practical course for engineers, makers and builders.

Oryx Institute Faculty of Applied Engineering



Oryx Institute Press · Windhoek, Namibia

FIRST EDITION

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A practical course for engineers, makers and builders.

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PREFACE

Preface

Why this book exists.

This book is for people who work with their hands and their heads. It teaches measurement as a habit rather than a formula: define, measure, record, verify, report.

Each chapter opens with a short list of outcomes, builds the idea in small sections, and closes with a summary, key terms and review questions. Work through the examples at your bench, not only on the page.

We thank the partners and learners whose problems shaped these chapters.

Tangi Iigonda
PRINCIPAL

CHAPTER 1

01

Why we measure

“If you cannot measure it, you cannot repeat it.”

IN THIS CHAPTER

1.1 Numbers in the trades 1.2 The measurement cycle 1.3 Units and symbols 1.4 Keeping a record

What you will be able to do

- Explain why a trade depends on repeatable numbers.
- Follow the five-step measurement cycle.
- Use the base units correctly.
- Keep a record that another person can follow.

1.1 Numbers in the trades

Every trade begins with a number. A carpenter cuts to a length, an electrician checks a voltage, a nurse reads a dose. When the number is wrong, the work is wrong, however good the hands that did it.

1.2 The measurement cycle

Measuring well is a habit with five steps. First, define exactly what you need to know. Then measure it with a suitable instrument, record the result at once, verify it against a second reading, and report it clearly.

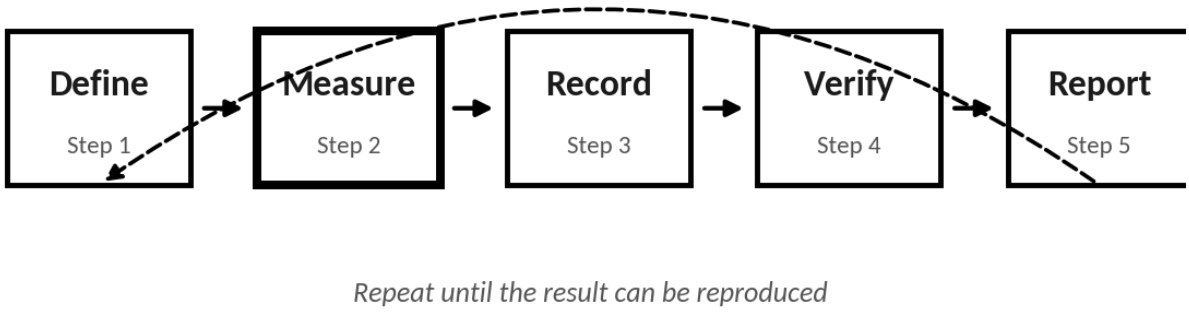


Figure 1.1. The measurement cycle.

> **WORKED EXAMPLE**
> A bench is measured three times: 1,200 mm, 1,201 mm and 1,199 mm. The mean is 1,200 mm, and the readings agree within 1 mm.

1.3 Units and symbols

Quantity	Unit	Symbol
Length	metre	m
Mass	kilogram	kg
Time	second	s
Electric current	ampere	A

1.4 Keeping a record

A result you did not write down is a result you do not have. Record the date, the instrument, the reading and your initials. Never rub out an entry; strike it through and write the correction beside it.

Chapter summary

- Measurement turns a question into a number you can repeat.
- The cycle has five steps: define, measure, record, verify, report.
- Always record the date, instrument, reading and initials.

Key terms

Term	Meaning
Precision	How closely repeated measurements agree.
Accuracy	How close a measurement is to the true value.
Verify	Check a result against a second, independent reading.

Review questions

1. Name the five steps of the measurement cycle.

2. Why must a result be recorded at the moment it is taken?

CHAPTER 2

Units and standards

This chapter is a placeholder. It follows the same structure as Chapter 1.

IN THIS CHAPTER

2.1 The base units 2.2 Prefixes 2.3 Standards and traceability

APPENDIX A

Worked solutions

Chapter 1, question 1
Define, measure, record, verify, report.

Chapter 1, question 2
Because memory fails, and a result that is not recorded cannot be repeated or checked.

GLOSSARY

Glossary

Term	Definition
Accuracy	Closeness of a measurement to the true value.
Calibration	Comparing an instrument with a known standard.
Precision	Agreement between repeated measurements.
Uncertainty	The doubt that remains in a measured value.

INDEX

Index

PLACEHOLDER

Generate the index in Word: mark entries, then References > Insert Index.



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